

Total No. of Questions : 8]

SEAT No. :

PC2421

[Total No. of Pages : 2

[6354]-541

B.E. (E&TC)

FIBER OPTIC COMMUNICATION
(2019 Pattern) (Semester - VIII) (404190)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8
- 2) Figures to the right indicate full marks.
- 3) Assume suitable data if necessary.

- Q1)** a) What are the requirements of material selection for photodetectors. [5]
b) Draw the structure of APD and explain its working. List its advantages and drawbacks. [6]
c) Explain different types of noise in the receiver. [6]

OR

- Q2)** a) Draw structure of p-i-n photodiode and explain its operation in brief. Plot the responsivity curve as function of wavelength for p-i-n photodiodes. [6]
b) A given APD has a quantum efficiency of 65% at a wavelength of 900 nm. Suppose 0.5 micro watt of optical power produces a multiplied photocurrent of 10 micro A. Find multiplication factor. [6]
c) Explain receiver structure in detail. [5]
- Q3)** a) Let the data rate of 20mb/s and BER of 10^{-9} . For the receiver PIN photodiode operating at 850 nm, the required receiver input signal is -42dBm. The LED is used as a source can couple 50uW (-13dBm) average optical power into a fiber flylead with a 50 um core diameter. Assume 1dB loss occurs when the fiber flylead is connected to the cable and another 1dB connector loss at the cable-photo detector interface. System margin of 6dB. Let attenuation per km is 3.5dB/km. Estimate link length. [7]
b) Write short note on optical coupler, circulator and isolator. [6]
c) Illustrate fiber grating with the help of a neat diagram. [5]

OR

P.T.O.

- Q4)** a) An optical fiber transmission system is to be designed to operate on 8km length without repeaters. The rise time of the chosen components are source LED : 8ns, Fiber cable: intermodal : 5ns/km, Intramodal: 1ns/km, Detector PIN :6ns Estimate maximum bit rate that may be achieved on the link when using NRZ and RZ formats. [6]
- b) Enlist types of optical amplifiers.Explain EDFA in detail. [6]
- c) Illustrate the significance of WDM components with suitable block diagram. List the formulas for fused-Biconical coupler. [6]

- Q5)** a) Define network topology. State and explain types of network topologies with suitable diagrams. [5]
- b) What is Fiber Distributed Data Interface (FDDI) and its features?Explain with diagram FDDI with respect to ring structure, Reference Model, frame and topology used. [8]
- c) Write short note on Active Optical network and Passive optical network.[5]

OR

- Q6)** a) Define Optical Network. Explain the term optical node & light path with a suitable diagram relative to optical network. [5]
- b) What is FTTX? What are the different categories of FTTX?Explain FTTX with respect to architecture. [6]
- c) Explain SONET with the help of structure,state advantages and applications.[7]

- Q7)** a) Describe OTDR with the help of a block diagram. Explain a typical trace of losses on OTDR. [7]
- b) Explain intermodal dispersion measurement technique. [5]
- c) Draw and explain EYE diagram measurement set up. [5]

OR

- Q8)** a) What is attenuation?Describe with necessary experimental arrangement for the measurement of the signal attenuation in an optical fiber by cutback technique. [7]
- b) Consider a long optical fiber with core refractive index $n_1=1.460$.Suppose that an engineer uses OTDR to locate a break in the fiber.If the break is located 15km away,What is the return time of an OTDR test pulse? [4]
- c) List the widely used optical measuring instruments and explain any three measuring instruments in detail. [6]

